

Yield Curve Equations for M-Library

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1 USD Market

1.1 Swap Equation

- 3M Libor Swap:

$$S_{4n}^{3ML} \sum_{i=1}^{2n} \tau_{2(i-1)}^{30/360} DF_{2i} = \sum_{i=1}^{4n} L_{i-1}^{3M} \tau_{i-1}^{ACT/360} DF_i \quad (1)$$

- 3M-6M Basis Swap:

$$\sum_{i=1}^{2n} \left[\prod_{j=1}^2 \{1 + (L_{2i-3+j}^{3M} + B_{4n}^{36}) \tau_{2i-3+j}^{ACT/360}\} - 1 \right] DF_{2i} = \sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/360} DF_{2i} \quad (2)$$

- Overnight Index Swap($\leq 2Y$):

$$S_{4n}^{OIS} \sum_{i=1}^n \tau_{4(i-1)}^{ACT/360} DF_{4i} = \sum_{i=1}^n O_{4(i-1)}^{Cmpd} \tau_{4(i-1)}^{ACT/360} DF_{4i} \quad (3)$$

- Libor-OIS Basis Swap($> 2Y$):

$$\sum_{i=1}^{4n} (O_{i-1}^{Avg} + B_{4n}^{LO}) \tau_{i-1}^{ACT/360} DF_i = \sum_{i=1}^{4n} L_{i-1}^{3M} \tau_{i-1}^{ACT/360} DF_i \quad (4)$$

⁰(Note) All discount factors are from OIS curve.

1.2 Par Rate Equation

(Note) In terms of calibration, par rate is the target function and $\boxed{X}$ is the explanatory variables.

- 3M Libor Swap:

$$S_{4n}^{3ML} = \frac{\sum_{i=1}^{4n} \boxed{L_{i-1}^{3M}} \tau_{i-1}^{ACT/360} DF_i}{\sum_{i=1}^{2n} \tau_{2(i-1)}^{30/360} DF_{2i}} \quad (5)$$

- 3M-6M Basis Swap: (No Analytical Solution)
- Overnight Index Swap($\leq 2Y$):

$$S_{4n}^{OIS} = \frac{\sum_{i=1}^n \boxed{O_{4(i-1)}^{Cmpd}} \tau_{4(i-1)}^{ACT/360} \boxed{DF_{4i}}}{\sum_{i=1}^n \tau_{4(i-1)}^{ACT/360} \boxed{DF_{4i}}} \quad (6)$$

- Libor-OIS Basis Swap($> 2Y$):

$$B_{4n}^{LO} = \frac{S_{2n}^{3ML} \sum_{i=1}^{2n} \tau_{2(i-1)}^{30/360} \boxed{DF_{2i}} - \sum_{i=1}^{4n} \boxed{O_{i-1}^{Avg}} \tau_{i-1}^{ACT/360} \boxed{DF_i}}{\sum_{i=1}^{4n} \tau_{i-1}^{ACT/360} \boxed{DF_i}} \quad (7)$$

2 EUR Market

2.1 Swap Equation

- 6M Libor Swap:

$$S_{4n}^{6ML} \sum_{i=1}^n \tau_{4(i-1)}^{30/360} DF_{4i} = \sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/360} DF_{2i} \quad (8)$$

- 3M-6M Basis Swap¹:

$$\sum_{i=1}^{4n} (L_{i-1}^{3M} + B_{4n}^{36}) \tau_{i-1}^{ACT/360} DF_i = \sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/360} DF_{2i} \quad (9)$$

- Overnight Index Swap($\leq 2Y$):

$$S_{4n}^{OIS} \sum_{i=1}^n \tau_{4(i-1)}^{ACT/360} DF_{4i} = \sum_{i=1}^n O_{4(i-1)}^{Cmpd} \tau_{4(i-1)}^{ACT/360} DF_{4i} \quad (12)$$

- Libor-OIS Basis Swap($> 2Y$):

$$\sum_{i=1}^n (O_{4(i-1)}^{Avg} + B_{4n}^{LO}) \tau_{4(i-1)}^{ACT/360} DF_{4i} = \sum_{i=1}^{4n} L_{i-1}^{3M} \tau_{i-1}^{ACT/360} DF_i \quad (13)$$

2.2 Par Rate Equation

(Note) In terms of calibration, par rate is the target function and $\boxed{X}$ is the explanatory variables.

- 6M Libor Swap:

$$S_{4n}^{6ML} = \frac{\sum_{i=1}^{2n} \boxed{L_{2(i-1)}^{6M}} \tau_{2(i-1)}^{ACT/360} DF_{2i}}{\sum_{i=1}^n \tau_{4(i-1)}^{30/360} DF_{4i}} \quad (14)$$

¹It might be different from the major market convention, which equation is below.

$$\begin{cases} S_{4n}^{3ML} \sum_{i=1}^n \tau_{4(i-1)}^{30/360} DF_{4i} = \sum_{i=1}^{4n} L_{i-1}^{3M} \tau_{i-1}^{ACT/360} DF_i \\ (S_{4n}^{3ML} + B_{4n}^{36}) \sum_{i=1}^n \tau_{4(i-1)}^{30/360} DF_{4i} = \sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/360} DF_{2i} \end{cases} \quad (10)$$

$$\Leftrightarrow \sum_{i=1}^{4n} L_{i-1}^{3M} \tau_{i-1}^{ACT/360} DF_i + B_{4n}^{36} \sum_{i=1}^n \tau_{4(i-1)}^{30/360} DF_{4i} = \sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/360} DF_{2i} \quad (11)$$

- 3M-6M Basis Swap²:

$$B_{4n}^{36} = \frac{\sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/360} DF_{2i} - \sum_{i=1}^{4n} \boxed{L_{i-1}^{3M}} \tau_{i-1}^{ACT/360} DF_i}{\sum_{i=1}^{4n} \tau_{i-1}^{ACT/360} DF_i} \quad (15)$$

$$(or \quad B_{4n}^{36} = \frac{S_{4n}^{6ML} \sum_{i=1}^n \tau_{4(i-1)}^{30/360} DF_{4i} - \sum_{i=1}^{4n} \boxed{L_{i-1}^{3M}} \tau_{i-1}^{ACT/360} DF_i}{\sum_{i=1}^{4n} \tau_{i-1}^{ACT/360} DF_i}) \quad (16)$$

- Overnight Index Swap($\leq 2Y$):

$$S_{4n}^{OIS} = \frac{\sum_{i=1}^n \boxed{O_{4(i-1)}^{Cmpd}} \tau_{4(i-1)}^{ACT/360} \boxed{DF_{4i}}}{\sum_{i=1}^n \tau_{4(i-1)}^{ACT/360} \boxed{DF_{4i}}} \quad (20)$$

- Libor-OIS Basis Swap($> 2Y$):

$$B_{4n}^{LO} = \frac{S_{4n}^{3ML} \sum_{i=1}^n \tau_{4(i-1)}^{30/360} \boxed{DF_{4i}} - \sum_{i=1}^n \boxed{O_{4(i-1)}^{Avg}} \tau_{4(i-1)}^{ACT/360} \boxed{DF_{4i}}}{\sum_{i=1}^n \tau_{4(i-1)}^{ACT/360} \boxed{DF_{4i}}} \quad (21)$$

²It might be different from the major market convention, which equation is below.

$$B_{4n}^{36} = \frac{\sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/360} DF_{2i} - \sum_{i=1}^{4n} \boxed{L_{i-1}^{3M}} \tau_{i-1}^{ACT/360} DF_i}{\sum_{i=1}^n \tau_{4(i-1)}^{30/360} DF_{4i}} \quad (17)$$

$$(or \quad B_{4n}^{36} = \frac{S_{4n}^{6ML} \sum_{i=1}^n \tau_{4(i-1)}^{30/360} DF_{4i} - \sum_{i=1}^{4n} \boxed{L_{i-1}^{3M}} \tau_{i-1}^{ACT/360} DF_i}{\sum_{i=1}^n \tau_{4(i-1)}^{30/360} DF_{4i}}) \quad (18)$$

$$(or \quad B_{4n}^{LO} = S_{4n}^{6ML} - S_{4n}^{3ML}) \quad (19)$$

3 GBP Market

3.1 Swap Equation

- 6M Libor Swap:

$$S_{4n}^{6ML} \sum_{i=1}^{2n} \tau_{2(i-1)}^{ACT/365} DF_{2i} = \sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/365} DF_{2i} \quad (22)$$

- 3M-6M Basis Swap:

$$\sum_{i=1}^{4n} (L_{i-1}^{3M} + B_{4n}^{36}) \tau_{i-1}^{ACT/365} DF_i = \sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/365} DF_{2i} \quad (23)$$

- Overnight Index Swap:

$$S_{4n}^{OIS} \sum_{i=1}^n \tau_{4(i-1)}^{ACT/365} DF_{4i} = \sum_{i=1}^n O_{4(i-1)}^{Cmpd} \tau_{4(i-1)}^{ACT/365} DF_{4i} \quad (24)$$

3.2 Par Rate Equation

(Note) In terms of calibration, par rate is the target function and $\boxed{X}$ is the explanatory variables.

- 6M Libor Swap:

$$S_{4n}^{6ML} = \frac{\sum_{i=1}^{2n} \boxed{L_{2(i-1)}^{6M}} \tau_{2(i-1)}^{ACT/365} DF_{2i}}{\sum_{i=1}^{2n} \tau_{2(i-1)}^{ACT/365} DF_{2i}} \quad (25)$$

- 3M-6M Basis Swap:

$$B_{4n}^{36} = \frac{\sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/365} DF_{2i} - \sum_{i=1}^{4n} \boxed{L_{i-1}^{3M}} \tau_{i-1}^{ACT/365} DF_i}{\sum_{i=1}^{4n} \tau_{i-1}^{ACT/365} DF_i} \quad (26)$$

$$(or \quad B_{4n}^{36} = \frac{S_{4n}^{6ML} \sum_{i=1}^{2n} \tau_{2(i-1)}^{ACT/365} DF_{2i} - \sum_{i=1}^{4n} L_{i-1}^{3M} \tau_{i-1}^{ACT/365} DF_i}{\sum_{i=1}^{4n} \tau_{i-1}^{ACT/365} DF_i}) \quad (27)$$

- Overnight Index Swap:

$$S_{4n}^{OIS} = \frac{\sum_{i=1}^n \boxed{O_{4(i-1)}^{Cmpd}} \tau_{4(i-1)}^{ACT/365} \boxed{DF_{4i}}}{\sum_{i=1}^n \tau_{4(i-1)}^{ACT/365} \boxed{DF_{4i}}} \quad (28)$$

4 JPY Market

4.1 Swap Equation

- 6M Libor Swap:

$$S_{4n}^{6ML} \sum_{i=1}^{2n} \tau_{2(i-1)}^{ACT/365} DF_{2i} = \sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/360} DF_{2i} \quad (29)$$

- 3M-6M Basis Swap:

$$\sum_{i=1}^{4n} (L_{i-1}^{3M} + B_{4n}^{36}) \tau_{i-1}^{ACT/360} DF_i = \sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/360} DF_{2i} \quad (30)$$

- Overnight Index Swap:

$$S_{4n}^{OIS} \sum_{i=1}^n \tau_{4(i-1)}^{ACT/365} DF_{4i} = \sum_{i=1}^n O_{4(i-1)}^{Cmpd} \tau_{4(i-1)}^{ACT/365} DF_{4i} \quad (31)$$

4.2 Par Rate Equation

(Note) In terms of calibration, par rate is the target function and $\boxed{X}$ is the explanatory variables.

- 6M Libor Swap:

$$S_{4n}^{6ML} = \frac{\sum_{i=1}^{2n} \boxed{L_{2(i-1)}^{6M}} \tau_{2(i-1)}^{ACT/360} DF_{2i}}{\sum_{i=1}^{2n} \tau_{2(i-1)}^{ACT/365} DF_{2i}} \quad (32)$$

- 3M-6M Basis Swap:

$$B_{4n}^{36} = \frac{\sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/360} DF_{2i} - \sum_{i=1}^{4n} \boxed{L_{i-1}^{3M}} \tau_{i-1}^{ACT/360} DF_i}{\sum_{i=1}^{4n} \tau_{i-1}^{ACT/360} DF_i} \quad (33)$$

$$(or \quad B_{4n}^{36} = \frac{S_{4n}^{6ML} \sum_{i=1}^{2n} \tau_{2(i-1)}^{ACT/365} DF_{2i} - \sum_{i=1}^{4n} L_{i-1}^{3M} \tau_{i-1}^{ACT/360} DF_i}{\sum_{i=1}^{4n} \tau_{i-1}^{ACT/360} DF_i}) \quad (34)$$

- Overnight Index Swap:

$$S_{4n}^{OIS} = \frac{\sum_{i=1}^n \boxed{O_{4(i-1)}^{Cmpd}} \tau_{4(i-1)}^{ACT/365} \boxed{DF_{4i}}}{\sum_{i=1}^n \tau_{4(i-1)}^{ACT/365} \boxed{DF_{4i}}} \quad (35)$$